

WayOfMono — Technical Overview

System Architecture

WayOfMono is built on an Interface-Agnostic Architecture — core logic and tools are shared across all major AI coding agent platforms.

Four-Layer Architecture

Layer 1: Agent Frontends (7 tools)

- **wocode** — High-performance coding assistant CLI (our flagship)
- **wouser** — General-purpose user agent SDK + CLI
- **Claude Code** — Anthropic's coding agent
- **OpenCode** — Open-source TUI-driven agent
- **Gemini CLI** — Google's multimodal agent
- **Pi** — Pi Agent standard

Key Technical Differentiators

1. Multi-Provider LLM Abstraction

30+ LLM providers under a single unified API. Supports streaming, tool calling, thinking budgets, and OAuth. Providers include OpenAI, Anthropic, Google Gemini, AWS Bedrock, Mistral, Azure, DeepSeek, xAI, Groq, and more.

2. Custom TUI Framework

Built from scratch (not based on Ink/React/Blessed). Features differential rendering, Kitty image protocol, markdown rendering, fuzzy finding, and full keyboard state machine. 11,160 lines, 12 components.

3. Context Compaction Engine

Branch summarization and token estimation to manage LLM context windows across

Technology Stack

Layer	Technology
Language	TypeScript (ES2024, NodeNext)
Runtimes	Node.js 22+, Deno 2.x, Bun
Package Manager	pnpm 10 workspaces
Build	tsc (TypeScript 6.x)
LLM Providers	30+ (OpenAI, Anthropic, Gemini, Bedrock, etc.)
Database	SQLite (Prisma ORM)
Frontend	Next.js 16, React 19, Tailwind CSS 4
Container	Podman (Quadlets) / Docker
Reverse Proxy	Caddy 2
CI/CD	GitHub Actions
Versioning	Changesets

Deployment

- **Docker multi-stage build:** Node 22 Alpine, Prisma migration, Next.js standalone output
- **Podman Quadlets:** Systemd-managed containers with Caddy reverse proxy
- **Cloudflare Tunnel:** Optional for zero-config HTTPS access
- **Devbox environment:** Reproducible development with Node 22, Deno, Bun, Podman pre-configured